

Assignment - 4

u)

$$f(A, B, C) = \sum m(1, 2, 5, 7)$$

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

And need

$$B \cdot A = 1$$

$$B \cdot C = 1$$

$$B \cdot C = 1$$

05 need

$$F = B \cdot A + B \cdot C + A \cdot C$$

$$F_1 = \sum m(1, 2, 5, 7)$$

$$F_2 = \sum m(1, 2, 5, 7)$$

$$F_3 = \sum m(1, 2, 5, 7)$$

$$F_4 = \sum m(1, 2, 5, 7)$$

$$F_5 = \sum m(1, 2, 5, 7)$$

Answer

(2)

$$F(A, B, C) = AB + AC + BC$$

AND array

$$P_1 = AB$$

$$P_2 = AC$$

$$P_3 = BC$$

OR array

$$F = P_1 + P_2 + P_3$$

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$(3) F_1 = \sum m(1, 3, 5, 7) = C$$

$$F_2 = \sum m(0, 2, 6, 7) = AB + A'C$$

$$P_1 = C \rightarrow F_1$$

$$P_2 = AB \rightarrow F_2$$

$$P_3 = A'C \rightarrow F_2$$

(4)
~~4~~
F₁

(4) $F_1 = \sum m(1, 2, 3, 5, 7)$
 $F_2 = \sum m(0, 3, 6)$
 $F_3 = \sum m(0, 2, 4, 6)$

Simplified

$$F_1 = A'B + A'C + BC$$
$$F_2 = A'B'C' + A'BC + AB'C'$$
$$F_3 = C'$$